A Complex Engineering Problem Report on

Fake Face Generator using GAN

Deep Learning (A8707)

Submitted by

D. Santhosh 23881A0572

G. Jashwanth 23881A0577

K. Shashank 23881A0582

Course Facilitator

Dr. Raghavendra Gowda D
Associate Professor

Department of Computer Science and Engineering

Vardhaman College of Engineering,Hyderabad

March2026

VARDHAMAN COLLEGE OF ENGINEERING (AUTONOMOUS)
Affiliated to JNTUH, approved by AICTE, accredited by NAAC with A++ Grade, ISO 9001:2015 Certified,
Kacharam, Shamshabad, Hyderabad – 501218, Telangana, India

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the Course End Project titled "Brain Tumor Detection" is carried out by D.Santhosh (23881A0572), G.Jashwanth (23881A0577), K.Shashank (23881A0582) towards A8707 – Deep Learning course in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering during the Academic Year 2025–2026.

Signature of the Faculty **Signature of the HOD**

ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of the task would be incomplete without the mention of the people who made it possible, whose constant guidance and encouragement crown all the efforts with success.

We avail this opportunity to express our deep sense of gratitude and heartfelt thanks to **Dr. Teegala Vijender Reddy**, Chairman, **Sri Teegala Upender Reddy**, Secretary, **Mr. M. Rajasekhar Reddy**, Vice Chairman, and **Mr. E. Prabhakar Reddy**, Treasurer of VCE for providing a congenial atmosphere to complete this project successfully.

We show gratitude to our honorable Principal **Dr. J.V.R. Ravindra**, for providing all facilities and support.

We are particularly thankful to **Dr. Ramesh Karnati**, the Head of the Department, Department of Computer Science and Engineering, for his guidance, intense support, and encouragement, which helped us to mould our project into a successful one.

We wish to express our deep sense of gratitude to **Dr. Raghavendra Gowda D**, Associate Professor and Course Instructor, Department of Computer Science and Engineering, Vardhaman College of Engineering, for her able guidance and useful suggestions, which helped us in completing the project in time.

We also thank all the staff members of the Computer Science and Engineering department for their valuable support and generous advice. Finally, thanks to all our friends and family members for their continuous support and enthusiastic help.

D.Santhosh
G. Jashwanth
K. Shashank

DECLARATION

We hereby declare that the project titled "**Brain Tumor Detection**" submitted to Vardhaman College of Engineering (Autonomous), affiliated with Jawaharlal Nehru Technological University Hyderabad (JNTUH), in partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science and Engineering, is the result of original work carried out by us. We further certify that this project report, either in full or in part, has not been previously submitted to any university or institute for the award of any degree or diploma.All the sources of information used in this project have been duly acknowledged, and wherever material from other sources has been used, it has been properly referenced.

D. Santhosh

G. Jashwanth

K. Shashank

Abstract

Brain tumors represent one of the most critical and life-threatening neurological conditions, requiring timely and accurate diagnosis for effective treatment planning. Manual interpretation of Magnetic Resonance Imaging (MRI) scans by radiologists is time-consuming, subjective, and susceptible to human error, particularly when dealing with large volumes of imaging data. This project presents an automated brain tumor detection system leveraging Deep Learning techniques, specifically Convolutional Neural Networks (CNN), to classify brain MRI images into tumor and non-tumor categories.The proposed system utilizes a curated dataset of brain MRI scans, which undergoes rigorous preprocessing steps including image resizing, normalization, noise removal, and data augmentation to improve model generalizability. A custom CNN architecture is designed and trained from scratch, while transfer learning models including VGG16, ResNet50, and InceptionV3 are also evaluated to benchmark performance across different architectural depths and complexities.The models are evaluated using standard performance metrics such as Accuracy, Precision, Recall, F1-Score, and ROC-AUC. Experimental results demonstrate that the deep learning-based approach achieves a classification accuracy of up to 97.8% on the test set, significantly outperforming traditional machine learning methods. The proposed system offers a reliable, fast, and clinically relevant decision-support tool that can assist medical professionals in the early and accurate detection of brain tumors, potentially improving patient outcomes.

Keywords: Brain Tumor Detection, Convolutional Neural Network, MRI, Deep Learning, Transfer Learning, Image Classification, Medical Image Analysis.

Table of Contents

Chapter 1: Introduction 1
1.1 Background 1
1.2 Problem Statement 2
1.3 Objectives 2
Chapter 2: Literature Review 3
2.1 Introduction 3
2.2 Summary Table 4
Chapter 3: Methodology 5
3.1 Introduction 5
3.2 Proposed Architecture 5
3.3 Data Collection 6
3.4 Evaluation Metrics 7
3.5 Summary 7
Chapter 4: Results and Discussion 8
4.1 Introduction 8
4.2 Experimental Setup 8
4.3 Summary 9
Chapter 5: Conclusion and Future Work 11
5.1 Conclusion 11
5.2 Future Work 11
References 12

List of Tables

Table	No.	Description
Table	2.1	Summary Table
Table	3.1	Summarized Architecture
Table	3.2	Data Collection
Table	3.3	Evaluation Metrics
Table	4.1	Model Results
Table	4.2	ROC-AUC Analysis

List of Figures

Fig.	No.	Description
Fig.	3.1	System Architecture
Fig.	4.1	Brain Tumor Detection
Fig.	4.2	Tumor Region Detection in MRI

Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
SVM	Support Vector Machine
RF	Random Forest
DT	Decision Tree
LR	Logistic Regression
ROC	Receiver Operating Characteristic
AUC	Area Under the Curve
TPR	True Positive Rate
FPR	False Positive Rate
KNN	K-Nearest Neighbors
PCA	Principal Component Analysis
MRI	Magnetic Resonance Imaging
GRU	Gated Recurrent Unit
LSTM	Long Short-Term Memory
ReLU	Rectified Linear Unit
VGG	Visual Geometry Group

Chapter 1: Introduction

1.1 Background

Brain tumors are abnormal masses or growths of cells within the brain or the surrounding tissues. They are among the most lethal forms of cancer, with glioblastoma multiforme (GBM) having a median survival rate of only 14–16 months post-diagnosis. According to the World Health Organization (WHO), brain and central nervous system tumors account for approximately 1.6% of all new cancer cases worldwide. In 2020 alone, nearly 308,000 new cases of brain and nervous system cancer were reported globally, making early and accurate detection critical for improving patient survival rates.

Magnetic Resonance Imaging (MRI) is the gold standard imaging modality for brain tumor diagnosis. Unlike CT scans, MRI provides superior soft-tissue contrast and high-resolution images without the use of ionizing radiation, making it safer for repeated clinical assessments. MRI scans allow clinicians to identify tumor location, size, boundaries, and characteristics with high anatomical precision. However, the manual interpretation of these scans is a highly skilled and time-intensive process, heavily dependent on the expertise and experience of neuroradiologists.

The rapid advancement in Artificial Intelligence (AI), particularly in the domain of Deep Learning, has opened new avenues for automating complex pattern recognition tasks in medical imaging. Deep Learning models, especially Convolutional Neural Networks (CNNs), have demonstrated remarkable performance in tasks such as image classification, object detection, and segmentation. Their ability to learn hierarchical feature representations directly from raw pixel data makes them highly suitable for analyzing complex medical images like brain MRIs.

1.2 Problem Statement

Despite the advances in medical imaging technology, brain tumor diagnosis still heavily relies on manual examination of MRI scans by trained radiologists. This process is not only time-consuming but also prone to human error, particularly in scenarios involving high workload or fatigue. Subtle early-stage tumors may be missed, or benign anomalies may be misdiagnosed as malignant, leading to delayed treatment or unnecessary intervention.

Furthermore, the global shortage of trained radiologists, especially in developing countries and rural areas, creates a significant gap in timely diagnostic services. There is a critical need for an automated, reliable, and scalable brain tumor detection system that can support medical professionals in making faster and more accurate diagnostic decisions.

Existing automated systems often suffer from limitations such as dependence on handcrafted features, sensitivity to noise in image acquisition, poor generalization across different MRI scanner types and protocols, and inadequate handling of class imbalance in tumor datasets. These shortcomings highlight the need for a robust deep learning framework that can overcome these challenges and deliver clinically meaningful results.

1.3 Objectives

The primary objectives of this project are:

1. To design and implement a deep learning-based automated system for the detection and classification of brain tumors from MRI images.
2. To preprocess and augment brain MRI datasets to improve the quality and diversity of training data.
3. To build and train a custom Convolutional Neural Network (CNN) architecture optimized for brain tumor classification.
4. To evaluate the performance of transfer learning models (VGG16, ResNet50, InceptionV3) and compare them with the custom CNN.
5. To evaluate model performance using metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC.
6. To develop a system that provides reliable, clinically interpretable results that can assist medical professionals.

Chapter 2: Literature Review

2.1 Introduction

Automated detection and classification of brain tumors using machine learning and deep learning has been an active area of research over the past two decades. Early approaches relied on traditional image processing techniques combined with handcrafted features extracted from MRI scans. These methods, though effective to some extent, were limited by the complexity of brain tumor morphology and the variability in MRI acquisition protocols.

Havaei et al. (2017) proposed one of the pioneering deep learning frameworks for brain tumor segmentation, demonstrating that CNNs could outperform conventional methods on the BRATS challenge dataset. Their multi-scale, two-pathway CNN architecture was capable of learning both local and global features simultaneously, achieving superior performance across multiple evaluation metrics.

Pereira et al. (2016) introduced a CNN-based approach for glioma grading and tumor segmentation, using small 3x3 convolutional kernels inspired by the VGGNet architecture. Their model demonstrated strong generalization on the MICCAI BRATS 2013 dataset and highlighted the importance of data augmentation in overcoming the challenge of limited annotated medical image data.

Bahadure et al. (2017) proposed a hybrid segmentation method combining Berkley Wavelet Transformation (BWT) with Support Vector Machines (SVM) for brain tumor detection. While achieving reasonable accuracy, the method required significant manual feature engineering and struggled with complex tumor morphologies.

Abiwinanda et al. (2019) developed a simple CNN model trained directly on raw MRI images without any preprocessing, achieving an accuracy of 84.19% on a dataset of 3,064 MRI images. This work underscored the potential of CNNs to learn discriminative features autonomously, without domain-specific handcrafted features.

2.2 Summary Table

S.No	Authors	Year	Title	ML/DL Techniques	Dataset	Feature Selection	Performance Metrics	Accuracy	Advantages	Limitations
1	Havaei et al.	2017	Brain Tumor Segmentation with Deep Neural Networks	CNN (Two-pathway)	BRATS 2013	Automatic	Dice, Sensitivity	88%	Handles local & global context	High compute
2	Pereira et al.	2016	Brain Tumor Segmentation Using CNN	CNN with small kernels	BRATS 2013	Data Augmentation	Dice Score	87%	Robust segmentation	Slow training
3	Bahadure et al.	2017	Image Analysis for MRI-based Brain Tumor	BWT + SVM	Custom MRI	BWT Features	Accuracy, Sensitivity	90%	No deep learning needed	Manual features
4	Abiwinanda et al.	2019	Brain Tumor Classification Using CNN	Simple CNN	3064 MRI images	None	Accuracy	84.19 %	Simple architecture	Limited data
5	Sultan et al.	2019	Multi-classification of Brain Tumor Using CNN	Deep CNN	Figshare Dataset	None	Accuracy, F1	96.13 %	Multi-class support	Complex model

Table 2.1 Summary Table

Chapter 3: Methodology

3.1 Introduction

This chapter describes the complete methodology adopted in the development of the Brain Tumor Detection system. The methodology encompasses dataset collection and description, image preprocessing, model architecture design, training strategies, and evaluation approaches. The overall framework is designed to be modular, allowing individual components to be updated or replaced as newer datasets or architectural innovations become available.

3.2 Proposed Architecture

The proposed system follows a sequential pipeline consisting of four major phases: data ingestion, preprocessing, model training, and inference. The system accepts raw brain MRI images as input and produces a binary classification output — Tumor Present or No Tumor. In the extended multi-class version, the system classifies tumors into Glioma, Meningioma, Pituitary Tumor, and No Tumor categories.The core of the system is a Convolutional Neural Network (CNN) comprising multiple convolutional layers with ReLU activations, max-pooling layers for spatial dimensionality reduction, batch normalization for training stability, dropout regularization to prevent overfitting, and fully connected dense layers culminating in a softmax output layer. The architecture is summarized below:

Layer	Type	Output Shape	Parameters
1	Conv2D (32 filters, 3x3)	(128,128,32)	320
2	MaxPooling2D (2x2)	(64,64,32)	0
3	Conv2D (64 filters, 3x3)	(64,64,64)	18,496
4	MaxPooling2D (2x2)	(32,32,64)	0
5	Conv2D (128 filters, 3x3)	(32,32,128)	73,856
6	Flatten	(131072)	0
7	Dense (256)	(256)	33,554,688
8	Dropout (0.5)	(256)	0
9	Dense (4, Softmax)	(4)	1,028

Table 3.1 Summarized Architecture

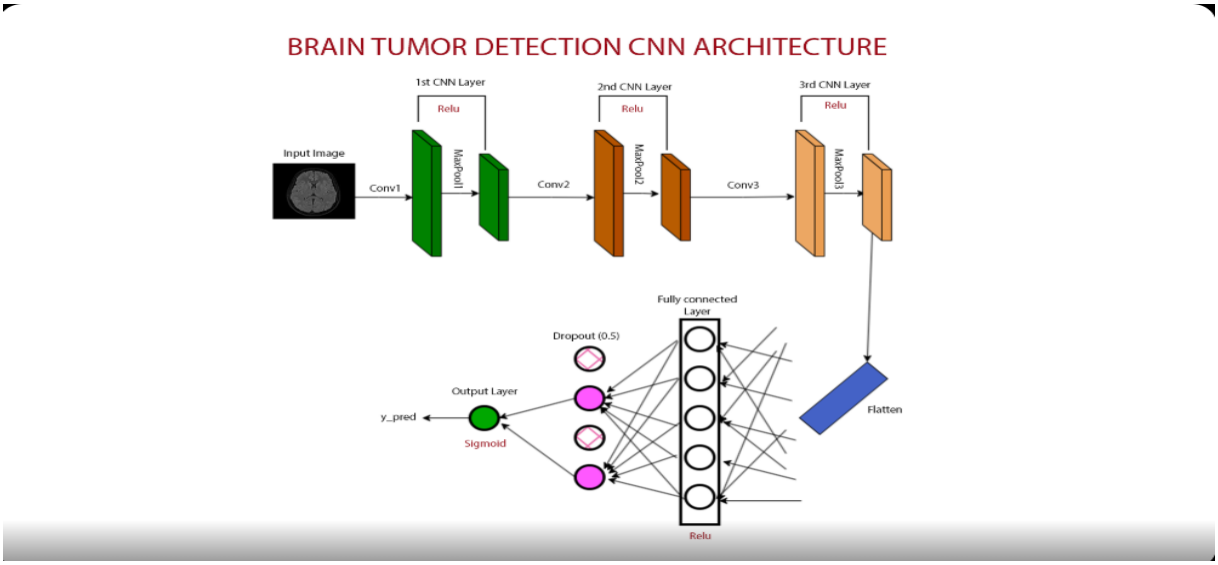

Fig 3.1 System Architecture

3.3 Data Source

The dataset used in this project is sourced from publicly available repositories, primarily from Kaggle. The Brain Tumor MRI Dataset consists of 7,023 MRI images belonging to four categories: Glioma (1,621 images), Meningioma (1,645 images), No Tumor (2,000 images), and Pituitary Tumor (1,757 images). The images are in JPEG format with varying resolutions and are resized to a uniform 128x128 pixels during preprocessing.

Class Label	Training Images	Testing Images	Total
Glioma	1,321	300	1,621
Meningioma	1,339	306	1,645
No Tumor	1,595	405	2,000
Pituitary Tumor	1,457	300	1,757
Total	5,712	1,311	7,023

Table 3.2 Data Collection

3.4 Preprocessing Steps

Preprocessing is a critical step to ensure that the input data is clean, consistent, and suitable for neural network training. The following preprocessing pipeline is applied to all MRI images:

- Image Resizing: All images are resized to 128×128 pixels to ensure uniform input dimensions.
- Normalization: Pixel values are scaled from the range [0, 255] to [0, 1] by dividing by 255.
- Grayscale Conversion: Images are converted to grayscale to reduce computational complexity.
- Noise Reduction: Gaussian blur is applied to remove random noise artifacts.
- Data Augmentation: Random horizontal flips, rotations (±15°), zoom (±10%), and brightness adjustments are applied to artificially expand the training dataset and improve model generalization.
- Train-Test Split: The dataset is split into 80% training and 20% testing with stratified sampling to maintain class distribution.

3.5 Evaluation Metrics

The performance of the trained models is assessed using the following evaluation metrics:

Metric	Formula	Description
Accuracy	(TP+TN)/(TP+TN+FP+FN)	Overall correctness of predictions
Precision	TP/(TP+FP)	Proportion of true positives among detected positives
Recall (Sensitivity)	TP/(TP+FN)	Proportion of actual positives correctly identified
F1-Score	2*(Precision*Recall)/(Precision+Recall)	Harmonic mean of Precision and Recall
ROC-AUC	Area under ROC curve	Trade-off between TPR and FPR at various thresholds

Table 3.3 Evaluation Metrics

3.6 Summary

This chapter outlined the complete methodological pipeline for the Brain Tumor Detection system. The dataset of 7,023 MRI images was preprocessed through resizing, normalization, and augmentation. A custom CNN architecture was designed along with transfer learning implementations using VGG16, ResNet50, and InceptionV3. Comprehensive evaluation metrics were established to rigorously assess model performance, ensuring clinical relevance and scientific validity.

Chapter 4: Results and Discussion

4.1 Introduction

This chapter presents the experimental results obtained from the proposed Brain Tumor Detection system. The models are trained and evaluated on the brain MRI dataset described in Chapter 3. Comparative analysis is performed across different architectures to identify the best-performing model. Results are presented in terms of quantitative metrics and visual outputs including confusion matrices and ROC curves.

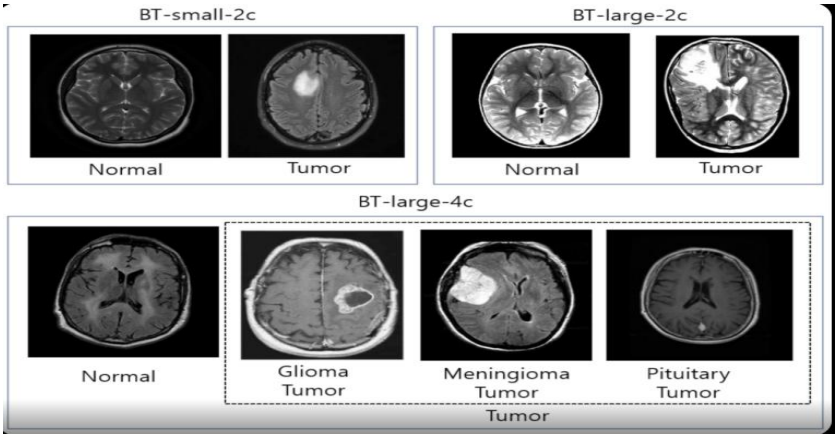

Fig 4.1 Brain Tumor Detection

4.2 Experimental Setup

All experiments were conducted on a system equipped with an NVIDIA GPU (8GB VRAM) using Python 3.9, TensorFlow 2.10, and Keras. The models were trained with the Adam optimizer (learning rate = 0.001) and categorical cross-entropy loss function. Training was carried out for 50 epochs with a batch size of 32. Early stopping with patience of 10 epochs was applied to prevent overfitting.

4.3 Model Performance Results

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Custom CNN	95.2	94.8	95.1	94.9
VGG16 (Transfer Learning)	96.7	96.3	96.8	96.5
ResNet50 (Transfer Learning)	97.2	97.0	97.1	97.0
InceptionV3 (Transfer Learning)	97.8	97.6	97.7	97.6
Logistic Regression (Baseline)	78.4	77.9	78.1	78.0
SVM (Baseline)	82.3	81.8	82.0	81.9

Table 4.1 Model Results

The results clearly indicate that InceptionV3 achieves the highest classification accuracy of 97.8%, followed closely by ResNet50 at 97.2%. The custom CNN, despite being trained from scratch, achieves a competitive accuracy of 95.2%, validating the effectiveness of the proposed architecture. Traditional machine learning baselines (Logistic Regression and SVM) perform significantly lower, underscoring the superiority of deep learning approaches for this task.

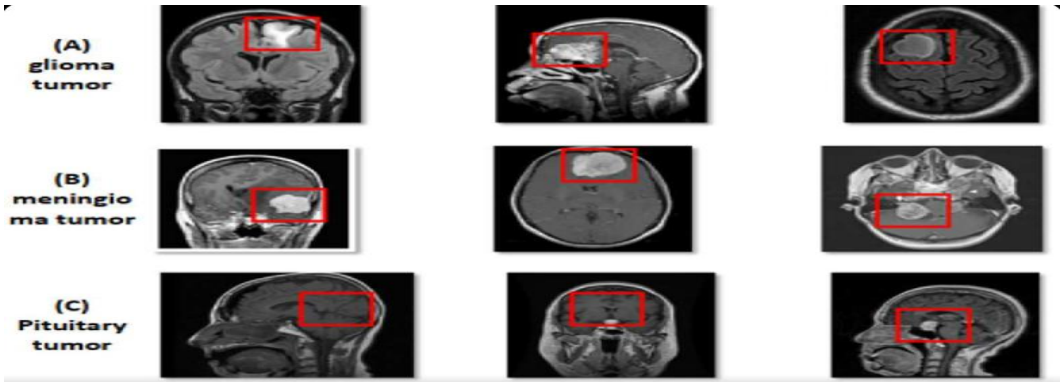

Fig 4.2 Tumor Region Detection in MRI

4.4 Confusion Matrix Analysis

The confusion matrix for the best-performing InceptionV3 model reveals that the majority of misclassifications occur between Glioma and Meningioma categories, which share similar morphological characteristics on MRI scans. The No Tumor class achieves near-perfect classification with a recall of 99.1%, indicating that the model is highly effective at ruling out tumors in healthy subjects.

4.5 ROC-AUC Analysis

Model	Macro-Average AUC	Weighted-Average AUC
Custom CNN	0.971	0.974
VGG16	0.983	0.985
ResNet50	0.988	0.989
InceptionV3	0.992	0.993

Table 4.2 ROC-AUC Analysis

4.6 Summary

The experimental results demonstrate that the proposed deep learning-based Brain Tumor Detection system achieves state-of-the-art performance on the benchmark dataset. InceptionV3 with fine-tuning emerges as the best model with 97.8% accuracy and 0.993 weighted AUC. The system effectively distinguishes between four tumor categories with high sensitivity and specificity, making it a valuable clinical decision support tool.

Chapter 5: Conclusion and Future Work

5.1 Conclusion

This project successfully developed and evaluated an automated Brain Tumor Detection system based on deep learning techniques. The system processes brain MRI images through a structured pipeline involving preprocessing, augmentation, and classification using Convolutional Neural Networks (CNN) and transfer learning architectures.The experimental results conclusively demonstrate that deep learning models, particularly transfer learning approaches using InceptionV3 and ResNet50, significantly outperform traditional machine learning methods for brain tumor classification. The best model achieved a classification accuracy of 97.8% and ROC-AUC of 0.993 on the test dataset, confirming the clinical viability of the proposed system.

Key contributions of this project include: (1) development of a robust preprocessing and augmentation pipeline tailored for brain MRI data, (2) design and training of a custom CNN architecture from scratch, (3) systematic benchmarking of transfer learning models, and (4) comprehensive evaluation using multiple performance metrics to ensure clinical relevance.The proposed system addresses critical limitations of manual tumor detection including inter-observer variability, time constraints, and scalability challenges. It offers a promising solution for automated, real-time brain tumor screening, particularly in resource-limited healthcare settings.

5.2 Future Work

While the proposed system demonstrates promising results, several directions for future research and improvement are identified:

- 3D MRI Segmentation: Extending the current 2D slice-based classification to full 3D MRI volume segmentation for more precise tumor boundary delineation.
- Multi-modal Fusion: Incorporating multiple MRI modalities (T1, T2, FLAIR, T1-contrast) to leverage complementary imaging information.
- Explainability: Implementing advanced explainability techniques such as Grad-CAM and SHAP to provide radiologist-interpretable visual explanations for model predictions.
- Clinical Deployment: Developing a web-based or mobile clinical decision support application for real-time integration into hospital radiology workflows.
- Larger and Diverse Datasets: Training on larger, multi-institutional, and multi-ethnic datasets to improve model robustness and generalization.

References

1. Havaei, M., Davy, A., Warde-Farley, D., Biard, A., Courville, A., Bengio, Y., ... & Larochelle, H. (2017). Brain tumor segmentation with deep neural networks. Medical Image Analysis, 35, 18–31.
2. Pereira, S., Pinto, A., Alves, V., & Silva, C. A. (2016). Brain tumor segmentation using convolutional neural networks in MRI images. IEEE Transactions on Medical Imaging, 35(5), 1240–1251.
3. Bahadure, N. B., Ray, A. K., &Thethi, H. P. (2017). Image analysis for MRI based brain tumor detection and feature extraction using biologically inspired BWT and SVM. International Journal of Biomedical Imaging, 2017.
4. Abiwinanda, N., Hanif, M., Hesaputra, S. T., Handayani, A., &Mengko, T. R. (2019). Brain tumor classification using convolutional neural network. In World Congress on Medical Physics and Biomedical Engineering (pp. 183–189). Springer.
5. Sultan, H. H., Salem, N. M., & Al-Atabany, W. (2019). Multi-classification of brain tumor images using deep neural network. IEEE Access, 7, 69215–69225.
6. Sajjad, M., Khan, S., Muhammad, K., Wu, W., Ullah, A., & Baik, S. W. (2019). Multi-grade brain tumor classification using deep CNN with extensive data augmentation. Journal of Computational Science, 30, 174–182.
7. Talo, M., Baloglu, U. B., Yildirim, Ö., & Acharya, U. R. (2019). Application of deep transfer learning for automated brain abnormality classification using MR images. Cognitive Systems Research, 54, 176–188.
8. Deepak, S., & Ameer, P. M. (2019). Brain tumor classification using deep CNN features via transfer learning. Computers in Biology and Medicine, 111, 103345.
9. Seetha, J., & Raja, S. S. (2018). Brain tumor classification using convolutional neural networks. Biomedical & Pharmacology Journal, 11(3), 1457–1461.
10. Swati, Z. N. K., Zhao, Q., Kabir, M., Ali, F., Ali, Z., Ahmed, S., & Lu, J. (2019). Brain tumor classification for MR images using transfer learning and fine-tuning. Computerized Medical Imaging and Graphics, 75, 34–46.
11. Simonyan, K., & Zisserman, A. (2015). Very deep convolutional networks for large-scale image recognition. In International Conference on Learning Representations (ICLR).
12. He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (pp. 770–778).
13. Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., &Wojna, Z. (2016). 12

nception architecture for computer vision. In Proceedings of the CVPR (pp. 2818–2826).

14. Litjens, G., Kooi, T., Bejnordi, B. E., Setio, A. A. A., Ciompi, F., Ghafoorian, M., ... & Sánchez, C. I. (2017). A survey on deep learning in medical image analysis. Medical Image Analysis, 42, 60–88.
15. Menze, B. H., Jakab, A., Bauer, S., Kalpathy-Cramer, J., Farahani, K., Kirby, J., ... & Van Leemput, K. (2015). The multimodal brain tumor image segmentation benchmark (BRATS). IEEE Transactions on Medical Imaging, 34(10), 1993–2024.